Assessing the Value of Gilead Sciences' Nine-Box Review

OIT 367 Final Project

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Yes, you may keep our project for future use.

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Executive Summary:

Gilead Sciences currently conducts two separate performance review processes each year: an annual performance review that assesses job performance during the evaluation period and a nine-box review that assesses both performance during the past year and potential for future contributions. The Gilead HR Analytics team recognizes that the nine-box review process is at least partially redundant with the annual performance review. Thus, they are interested in knowing whether they would lose valuable information if they were to drop the nine-box review from their evaluation process; if not, they can save time and effort for Gilead's managers by eliminating the nine-box review. The analyses in this project offer a view on the incremental value of the nine-box review as a means of informing this decision.

The value of the nine-box review was tested by assessing its ability to predict promotion outcomes in excess of the signal already contained in the annual performance review. This was tested across multiple models such as linear regression and gradient boosting machine (GBM) and against two response variables: number of promotions over two years after receiving the nine-box review, and promotion in the year immediately following the nine-box review. Although including the nine-box rating did appear to provide marginal increases in the accuracy of each predictive model, none of the differences in accuracy appeared large enough to be statistically significant. In other words, models with the nine-box rating were not sufficiently distinct from identical models built without the nine-box rating. Thus, our analysis fails to reject the null hypothesis that the nine-box evaluation holds no additional predictive power beyond the annual performance review data.

By not finding the nine-box incrementally predictive of promotion outcomes, this analysis suggests that the second review process is most likely redundant and could be dropped without depriving the HR Analytics team or wider company of unique information. However, the analysis cannot conclusively prove that the nine-box has no unique information: the metric could in fact be statistically significant for a different and/or larger set of employees, or over a different and/or larger timespan. This analysis also does not examine the business use of the nine-box evaluation beyond merely predicting promotions. Within the bounds of the question posed, however, the nine-box rating does not appear to add meaningful value in capturing employee quality (at least as measured by promotion outcomes).

Background Information:

Gilead Sciences currently conducts two separate performance review processes each year: an annual performance review and a nine-box review.

In the annual performance review, all employees are rated on a scale of 1 to 5 (1 = best, 5 = worst) with respect to their job performance during the evaluation period (the past year). Most employees receive a rating between 1 and 3, since Gilead works quickly and proactively (i.e., before the performance review cycle) to address employee performance problems that might lead to a rating in the bottom two buckets (4 and 5). Annual performance reviews typically take place in September of each year, and ratings from these reviews are a key driver of compensation.

After each year's annual performance review cycle is complete, a subset of employees are also assigned a rating as part of a nine-box review. This review involves a 3x3 matrix (i.e., nine boxes) along two axes: employee performance during the past year is rated on a scale of 1-3 (1 = best), while the employee's potential for future contributions to the company is rated on a scale of A-C (A = best). In practice, the performance rating in the nine-box review usually corresponds to the employee's rating from the annual performance review. The potential rating has been used in the past to determine which employees Gilead should target for specific investments in employee development.

Problem Statement and Questions:

The Gilead HR Analytics team recognizes that the nine-box review process is at least partially redundant with the annual performance review. Thus, they are interested in knowing whether they would lose any valuable data if they dropped the nine-box review from their evaluation process; if not, they can save time and effort for Gilead's managers by eliminating nine-box reviews.

The goal of our project is to analyze this question for Gilead, using response variables that capture an employee's promotion trajectory over time. In other words, we want to answer the following question:

> *If Gilead ignores the potential rating from the nine-box review, does an individual's rating in the annual performance review contain sufficient information to predict the employee's future promotion potential?*

As part of our deliverable, we considered a broad range of possible analyses to obtain further insights into promotions and employee performance, such as:

- How is the number of promotions impacted by performance reviews and ratings?

- How is the velocity of promotions impacted by performance reviews and ratings?
- Is it possible to accurately predict an employee's future promotions using only the employee's' annual performance review data (i.e., without the nine-box review data)?
- Does the relationship between promotions and performance ratings vary across functional areas, hire dates and compensation levels?

Company Stakeholders:

Our direct working team is the company's HR Analytics team. However, our question impacts Gilead's broader employee base, as the findings will influence the number of performance reviews each year.

Data Overview:

Our data was sourced directly from Gilead's HR Analytics team and contains key performance evaluation parameters for all current employees hired prior to September 2012 who received a nine-box rating during 2013. It includes both cross-sectional variables (e.g. employee functional area) as well as time-series variables (e.g. employee performance ratings for 2012-15). The most important predictor in our data set is the time-series variable: performance rating in the annual performance review. It serves as the key driver for our main response variable: promotion.

The initial file we received consists of a total of 1,002 rows denoting distinct employees and 14 columns. After our first discussion and based on our request, the Gilead team sent an updated file with three additional columns — the 2013 9 box performance rating, employee hire year and the employee function. Note that the updated file received from Gilead has only 996 observations.

Each employee data point contains:

- Unique employee ID
- Coded functional area (A to F)
- Hire year (1988 to 2012)
- Annual performance review for years 2012-15
    - Performance ratings (1 = exceptional to 5 = below expectations)
    - A z-score indicating distribution of performance rating across all Gilead employees. Note this distribution includes employees beyond our data set, such as individuals who are no longer with the company or those who were hired after 2012.
- Nine-Box Review for 2013 only
    - Performance rating (1 = highest performance to 3 = lower performance), which for the most part aligns with the results of the annual performance review

    - Future promotion rating (A = highest potential to C = lower potential)
- Most recent promotion date before the 2013 nine-box review
- Promotion dates since the 2013 nine-box review (maximum of 2)
- Grade or compensation level in 2013 (1 being an entry-level employee and 13 being the highest level), which serves as a proxy for seniority level

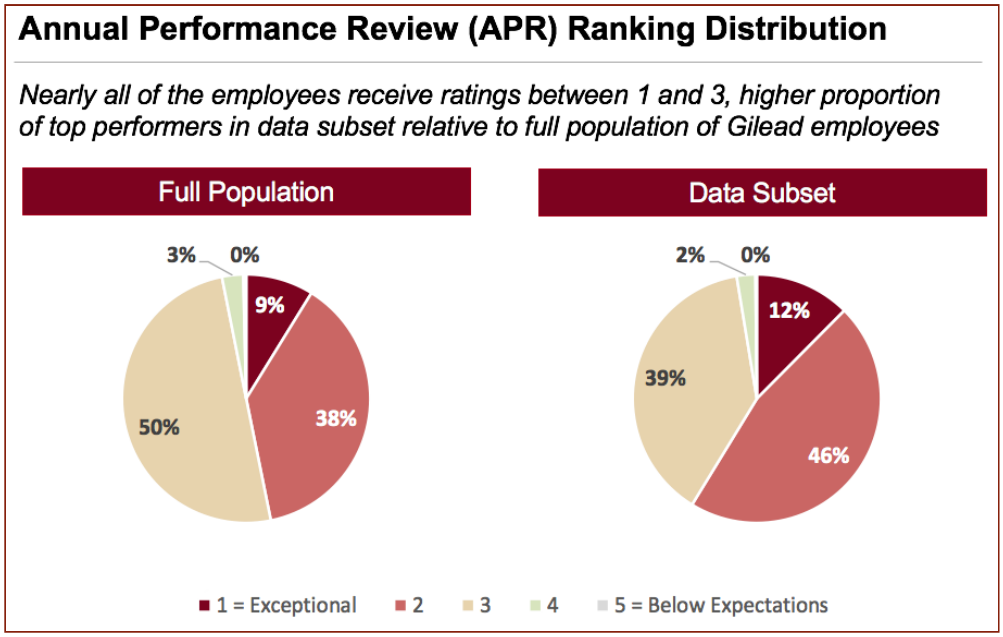
It is worthwhile to note the shortcoming of our hire year data. This information was used in calculating tenure, which subsequently served as a predictor in all of our models. Ideally, we would calculate tenure using the exact hire date. However, given concerns about employee privacy and our ability to potentially piece together exact individual identities from the information, hire year was given in place of exact hire date. As a result, we assumed a mid-year hire date for all employees (e.g. if the hire year was 2010, we assumed the hire date was 6/30/2010 in calculating tenure).

The HR Analytics group also provided a supplemental table with the overall performance rating distribution across years (see Appendix A).

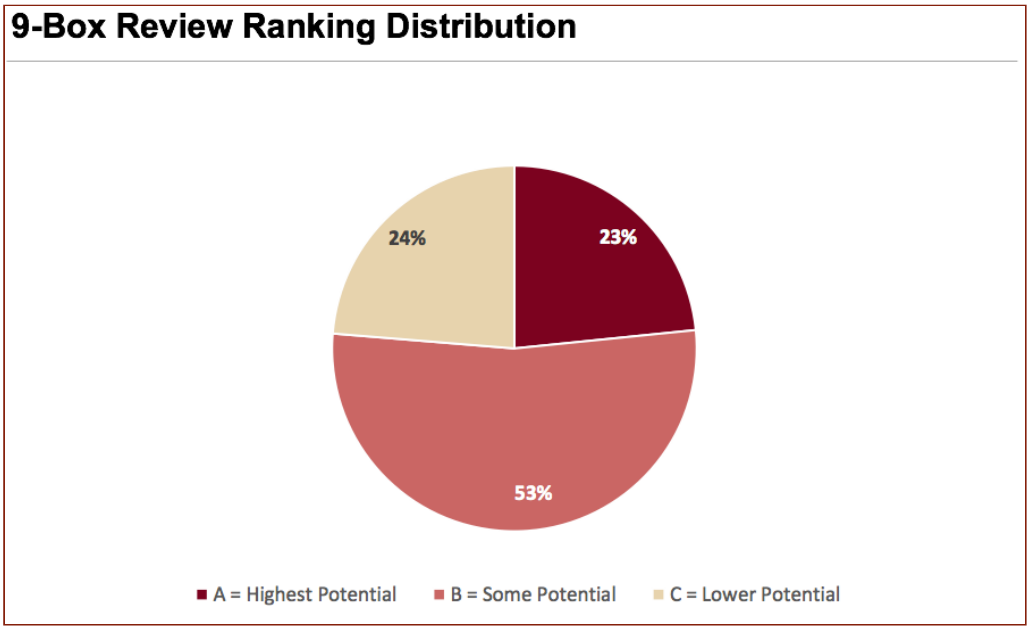
Analysis:

Before starting the analysis we took some steps to prepare the data and create additional variables. For example, we changed the employee grade variable to a factor rather than a numerical variable to address the fact that the scale is not linear (i.e., people spend unequal amounts of time and effort at each level) and some functions don't include certain levels. As part of our exploratory analysis we created bins for the employee grade variable to see if this would improve the model. Additionally, we created new variables such as the number of days to promotion as well as employee tenure, the latter to take into account that the likelihood of a promotion depends on the amount of time in the current role.

Using the z-scores provided to us, we conducted an analysis to determine how the employees in our data set compared to the overall employee population in terms of annual performance review ranking. We found that nearly all employees receive a rating between 1 and 3, and that our data set contains a higher proportion of top performance relative to the overall population, as shown in the figure below.



We also looked at the distribution of the nine-box review potential ratings, which is shown in the figure below. The majority of employees fall into the middle rating, and roughly equal number fall into the top and bottom rating.

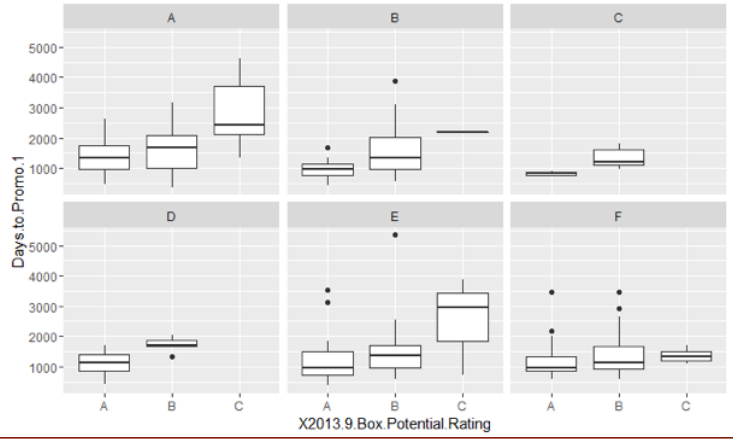


In order to visualize the relationship between the nine-box potential rating and promotion velocity, we plotted the number of days to promotion against the nine-box potential rating, as shown in the figure below.

Days to First Promotion

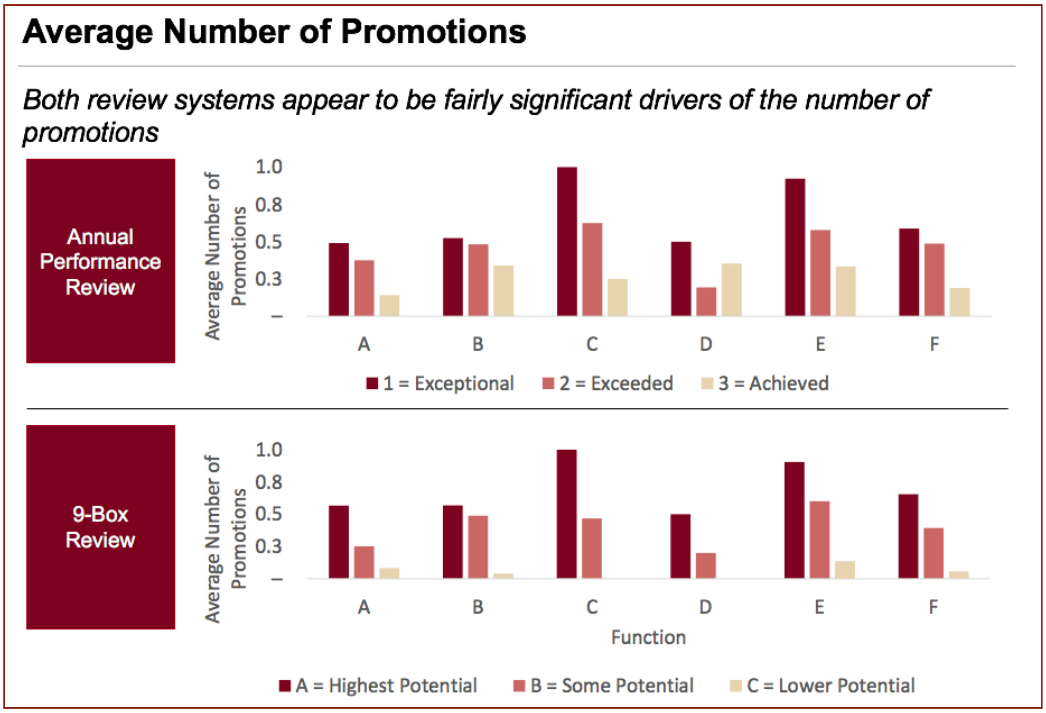
9-box rating looks to be a key driver in the number of days to promotion

Days to First Promotion by Function



The upward sloping trend in each plot indicates that the nine-box potential rating is positively correlated with the number of days to promotion in each function. The difference in slope in each plot points to the fact that the nine-box potential rating may be more valuable in predicting number of days to promotion for some functions than others. Relating this back to the objective of the project, the fact that the nine-box is valuable in predicting promotions does not mean that it is necessary. Our goal will be to determine whether the nine-box offers incremental predictive power, beyond the other predictors.

The next figure provides insight into the relationship between both the nine-box review and annual performance review and the number of promotions. The inference we draw is that both review systems appear to be strong predictors for the number of promotions.



After looking at these results, our next step was to use regression analysis to determine whether the predictive power offered by the nine-box review was significantly additive beyond the predictive power offered by the annual performance review and other signals. However, before we started the analysis we need to define our response variable precisely. We considered a few different approaches:

- Representing promotion as a binary variable for each promotion cycle
- Determining whether a promotion occurred the following year (2014) after the 2013 nine-box review cycle
- Counting the number of days to the next promotion after the 2013 nine-box review cycle
- Counting the number of promotions after 2013

A binary variable for promotion in each promotion cycle is complicated in that it requires multiple outputs—a yes or no response for each employee at each cycle. Furthermore, many promotions were granted off cycle. However, determining whether a promotion happened in 2014 for each individual resolves these complications, yielding a simple 1 or 0 as the response variable. Using the number of days between promotions was problematic in that many individuals have not been promoted since 2013, making it impossible to calculate a value for these individuals. Using the number of promotions as the response variable addresses this challenge by giving a zero to those who have not been promoted. As a result of this analysis and based on our discussion with Gilead, we decided to use whether a promotion occurred in 2014 and the number of promotions after receiving the 2013 nine-box review as our main response variables.

Prior to running the models, we randomized the data and created 20 folds on which we could perform cross-validation.

Exploration, defining null and alternative hypotheses, and variable selection

In our exploration of the questions we set out to answer in this project, we tried several different types of models including a linear regression and a gradient boosted machine tree-based model. For each model type, we performed 20-fold cross validation to build a model to predict the response variable for each employee based on performance ratings from 2012 and 2013, grade, tenure in role and function. Next, we performed 20-fold cross validation to build a model with all the same variables plus the nine-box rating. We would then compare evaluation metrics (RMSE or AUC depending on the response variable) for the model which did not include the nine-box rating with the model which did include the nine-box rating.

Since we applied 20-fold cross-validation to generate 20 RMSE or AUC values for each model, we were then able to use a t-test to determine whether the differences in the means of the two sets of 20 RMSE or AUC values were statistically significant. Specifically, we performed a one-sided Welch Two Sample t-test with the following null and alternative hypotheses, where μ represents the mean of the 20 RMSE or AUC values generated by 20-fold cross validation to test if adding the nine-box rating improves the accuracy of the model.

$$H_0: \mu^{RMSE}_{without\ 9B} - \mu^{RMSE}_{with\ 9B} = 0$$

$$H_1: \mu^{RMSE}_{without\ 9B} - \mu^{RMSE}_{with\ 9B} > 0$$

$$H_0: \mu^{AUC}_{without\ 9B} - \mu^{AUC}_{with\ 9B} = 0$$

$$H_1: \mu^{AUC}_{without\ 9B} - \mu^{AUC}_{with\ 9B} < 0$$

In all instances, the null hypothesis is that the difference in the means between the model without the nine-box rating and the model with the nine-box rating is zero. For RMSE, which decreases as model fit improves, the alternative hypothesis we selected is whether the difference in the means is greater than zero. For AUC, which increases as model fit improves, the alternative hypothesis we selected is whether the difference in the means is less than zero.

We had some discussion about whether or not to include the performance ratings for 2014 and 2015. We believed that excluding 2014 and 2015 ratings made sense both from a business as well as from a statistical perspective. In practice, the Gilead will not already have promotion information for future years when making present-day promotion decisions. The question for the HR Analytics team is given present-day information (which in this case is data from 2013 and years prior), who will get promoted? Furthermore from a statistical perspective, we learned that one of the common pitfalls of predictive analysis is using predictor variables that have direct information about the response variable. We do not want to peek into the future about 2014 and 2015 performance when building our model. For these reasons, performance data beyond the date of the 2013 nine-box review were excluded from our set of predictor variables.

Linear regression model
We ran a linear regression analysis initially because it is straightforward and easy to understand and explain. The linear model was designed to predict how many times each employee would be promoted (modeled as a continuous variable). Here we used RMSE to evaluate and compare the models.

Gradient boosted machine with Gaussian distribution
We also wanted to explore the possibility of interactions between the variables, such as an interaction between time in role and current job level. To address this we moved to a gradient boosted machine tree-based model, based on a Gaussian distribution, again predicting the number of promotions for each employee (modeled as a continuous variable). This model has the potential to capture interactions through the design of the tree, as splits in the branches of the tree are prioritized based on which combinations provide the most information to the model. The number of trees in the gradient boosted machine model was set to 5,000 trees after testing various alternatives. Here we also used RMSE to evaluate and compare the models.

Gradient boosted machine with Bernoulli distribution
Lastly, we ran a gradient boosted machine classification model, based on a Bernoulli distribution, predicting whether each employee would be promoted in 2014 (yes or no logistic regression) based on the same data. Building a logistic model on a slightly different response variable allowed us to view the problem from another angle to confirm or reject our earlier findings. Here we used AUC as the guiding metric to evaluate and compare the models, rather than RMSE.

As we explored ways to improve the models, we looked at including some different transformations of certain variables, such as the inverse of performance ratings (1/x), but saw deteriorations in RMSE as a result. Additionally, as mentioned above, we created bins for the employee grade variable to see if binning would improve the model, and similarly saw little to no effect. As a result, our final models do not involve transformations to the data set. We also concluded that LASSO regression would not be needed because the data set has 996 rows and very few predictors, even with interaction terms and higher powers. We also tried different values for number of folds, tree count, tree depth and learning rate (shrinkage). We saw the most improvement with learning rate and tree count, which informed the values in our final models.

Results:

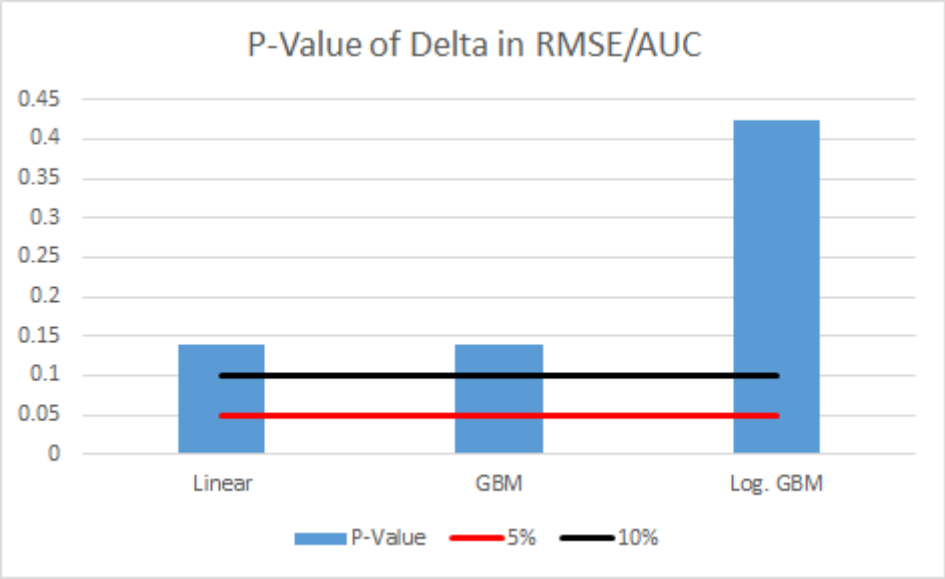
We did not find any statistically significant evidence that the nine-box review system adds any predictive power in determining promotions on top of the annual review data and other employee information, such as tenure, function, and grade controls. As a reminder, the quantity in question is not the accuracy of the predictive models, but rather the *change* in accuracy of a

model after including the nine-box review as compared to the baseline of the same models without the nine-box review data.

We used Residual Mean Squared Error, or RMSE -- a measure of the average dispersion between the model prediction and the actual outcome -- as an indicator of model accuracy. The more predictive a model, the smaller the distance between the model's prediction of number of promotions for each employee and the actual number of promotions received. For each model we used 20-fold cross validation to get 20 measurements of model accuracy. We averaged those 20 to calculate one RMSE for each model. For the logistic GBM model we used Area Under the Curve (AUC), rather than RMSE. AUC measures how much better than random allocation a model does in classifying predictors -- 0.5 is random chance and 1 is a perfectly predictive model.

For each pair of models with and without the inclusion of the nine-box rankings we then used a Welch Two-Sample t-test to identify if the difference between the model RMSE and AUC was statistically significant. Specifically, did the inclusion of the nine-box lower the RMSE or increase the AUC (improve the accuracy) of an otherwise identical model a statistically significant amount?

For each of the three models we tested, including the nine-box evaluation data did not statistically significantly increase the accuracy of the model. As the below chart shows, the smallest P-value was .14 (measuring the significance of adding the nine-box term to the linear model), still much larger than traditional cutoffs at the 1, 5, or 10% significance level and thus not statistically distinguishable from zero impact.



The mean RMSE and AUC, difference driven by including the nine-box data, and statistical significance of the difference for each of the three models is presented below. Note that

inclusion of the nine-box performance data did slightly improve the model accuracy (decrease RMSE and increase AUC) but never approaching statistical significance.

Model Accuracy						
	Linear	**Linear 9B**	**GBM**	**GBM 9B**	**Logistic**	**Logistic 9B**
Mean RMSE	0.501	0.482	0.5476	0.5198		
Mean AUC					0.8413	0.8459
P-Value		0.1401		0.1397		0.4246
Delta		-0.019		-0.0278		0.0046

Recommendations:

The data validates the intuition of the Gilead HR Analytics team. While the nine-box review data is highly correlated with promotion outcomes, it is not statistically predictive in excess of the information already gathered in the annual review process. A model predicting the timing of promotion for any given employee containing annual review data, tenure, function, and level is not enhanced by also including the nine-box review data. As long as promotion predictiveness is the (only) relevant business use for the nine-box data, the Gilead team should feel comfortable cutting the effort expended on the nine-box process and instead relying entirely on the standard annual review data.

A few potential conditions might justify further study. Firstly, the statistical tests failed to prove significance, they cannot prove insignificance. In the models tested, the nine-box rating did improve accuracy, albeit at a level so small it isn't statistically distinguishable from zero given the current body of data. It is possible that the true underlying effect is small but greater than zero -- something that could only be discovered via additional data (e.g. collection of nine-box for more years or more employees) and additional study. Whether or not a potential informational content detectable only across many thousands of datapoints is relevant enough to managerial decisions to justify further study is beyond the scope of this analysis, although such a scenario seems implausible.

Further, it is worth considering the low level of overall magnitude. The RMSE, depending on the exact model specifications, were around 0.5 in a situation where total promotions received ranged from 0 to 2 for each employee in the data studied. That implies some signal -- perhaps enough to separate top performers from those unlikely to receive a promotion, at least in most cases -- but clearly does not capture all of the data that goes into promotion decisions. It may be worth considering what other data or factors go into promotion decisions before cutting the nine-box review process. For instance, even though the data recorded doesn't show significance, if decision makers find the process of formulating the review impactful in promotion outcomes -- for example, by implicitly collecting other data not explicitly recorded -- then eliminating the process could prove detrimental.

Conclusion:

Gilead uses two review systems: an annual performance review and a nine-box review. The Gilead HR Analytics team asked us to investigate if the nine-box data contains any incremental value in predicting employee promotions beyond the data already captured in the annual review process.

We tested this question by building models to predict promotion outcomes utilizing only the annual performance review rating and other available HR data such as function, level, and tenure. We then tested if the accuracy of the models could be improved by using the nine-box rating as an additional predictor. In the interest of robustness, we used both a linear regression model and a Gaussian gradient boosting machine (GBM) model to predict the response variable "number of promotions in the two years after review" and additionally built a Bernoulli GBM model predict the response "will an employee get promoted in the next year".

In all cases adding the nine-box data did not yield additional predictive accuracy beyond what was already contained in the annual review performance rating. Specifically the change in RMSE and AUC when adding nine-box ranking to a model was not statistically distinguishable from zero for any model tested. Thus, based on the employees and years for which data is provided, the Gilead HR Analytics team should feel confident that the nine-box review did not provide any insight into employee promotion timing beyond what was already captured during the annual review process.

Appendix A: Performance Rating Distribution for Entire Gilead Sciences Employee Base

Performance Rating Distribution:

Year	Exceptional (1)	Exceeded (2)	Achieved (3)	Improvement Needed (4)	Below Expectations (5)
2015	694	3,023	3,744	191	22
2014	580	2,510	3,520	179	26
2013	495	2,076	2,811	127	15
2012	412	1,794	2,297	172	25

Appendix B: R Code for Analysis

```
rm(list=ls())
setwd("~/GSB/MBA1 - Q2 - Second Quarter/OIT 367 - Business Intelligence from Big Data/Final Project - Gilead")
#load the modified version of the OIT367 code with the updated values for learning rate = 0.05 and ntrees = 5000
source("~/GSB/MBA1 - Q2 - Second Quarter/OIT 367 - Business Intelligence from Big Data/OIT367_R_package.R")

#----------------------
#read data
data = read.csv("gild.csv")
head(data)

#data cleanup
data$Days.to.Promo.1 = NULL
data$Days.to.Promo.2 = NULL
data$X2014 = NULL
data$X2015 = NULL
data$X2013.Grade = as.factor(data$X2013.Grade)  #make the grade a factor (since non-linear with the jump in level)
data$Grade.Bin = as.factor(data$Grade.Bin)  #make the grade bin a factor
data$Function = as.factor(data$Function)  #make the function a factor
head(data)

#----------------------
#randomizing the data
n = nrow(data)
data.randomized = data[sample(n),]

#----------------------
#cross-validation and matrix setup
nFold = 20
rmseMatrix = matrix(0,nrow=nFold,ncol=8)

#----------------------
#Set the number of trees for GBM model
ntrees = 5000

#----------------------
#This model does not transform data set, excludes annual performance rankings from years 2014-15
```

```
flds = createFolds(seq(1,nrow(data.randomized)), nFold, list = TRUE, returnTrain = FALSE)
for (i in 1:nFold){
  ind = flds[[i]]
  data.training = data.randomized[-ind,]
  data.training.grade = data.training
  data.training.grade$Grade.Bin = NULL
  data.training.grade.bin = data.training
  data.training.grade.bin$X2013.Grade = NULL
  data.validation = data.randomized[ind,]
  data.validation.grade = data.validation
  data.validation.grade$Grade.Bin = NULL
  data.validation.grade.bin = data.validation
  data.validation.grade.bin$X2013.Grade = NULL
  #linear model without 9B using grade
  model.linear = lm(Number.of.Promotions
                ~ X2012+X2013
                +X2013.Grade+Function+Tenure.in.Role,
                data=data.training.grade)
  predictions.linear = predict(model.linear, newdata = data.validation.grade)
  validation.residuals.linear = data.validation.grade$Number.of.Promotions-predictions.linear
  validationRMSE.linear = sqrt(mean(validation.residuals.linear^2, na.rm = TRUE))
  rmseMatrix[i,1] = validationRMSE.linear
  #linear model without 9B using grade.bin
  model.linear = lm(Number.of.Promotions
                ~ X2012+X2013
                +Grade.Bin+Function+Tenure.in.Role,
                data=data.training.grade.bin)
  predictions.linear = predict(model.linear, newdata = data.validation.grade.bin)
  validation.residuals.linear = data.validation.grade.bin$Number.of.Promotions-predictions.linear
  validationRMSE.linear = sqrt(mean(validation.residuals.linear^2, na.rm = TRUE))
  rmseMatrix[i,2] = validationRMSE.linear
  #linear model with 9B using grade
  model.linear = lm(Number.of.Promotions
                ~ X2012+X2013
                +X2013.Grade+Function+Tenure.in.Role+X9B,
                data=data.training.grade)
  predictions.linear = predict(model.linear, newdata = data.validation.grade)
  validation.residuals.linear = data.validation.grade$Number.of.Promotions-predictions.linear
  validationRMSE.linear = sqrt(mean(validation.residuals.linear^2, na.rm = TRUE))
  rmseMatrix[i,3] = validationRMSE.linear
  #linear model with 9B using grade.bin
  model.linear = lm(Number.of.Promotions
                ~ X2012+X2013
```

```
            +Grade.Bin+Function+Tenure.in.Role+X9B,
            data=data.training.grade.bin)
predictions.linear = predict(model.linear, newdata = data.validation.grade.bin)
validation.residuals.linear = data.validation.grade.bin$Number.of.Promotions-predictions.linear
validationRMSE.linear = sqrt(mean(validation.residuals.linear^2, na.rm = TRUE))
rmseMatrix[i,4] = validationRMSE.linear
#GBM model with Gaussian without 9B using grade
model.gbm = gbm(Number.of.Promotions
            ~X2012+X2013
            +X2013.Grade+Function+Tenure.in.Role,
            data = data.training.grade,
            distribution="gaussian",
            n.trees=ntrees,
            shrinkage=0.05,
            interaction.depth=2,
            bag.fraction = 0.5,
            train.fraction = 0.5,
            cv.folds = 3)
predictions.gbm = predict(model.gbm, newdata = data.validation.grade, n.trees=ntrees)
validation.residuals.gbm = data.validation.grade$Number.of.Promotions-predictions.gbm
validationRMSE.gbm = sqrt(mean(validation.residuals.gbm^2, na.rm = TRUE))
rmseMatrix[i,5] = validationRMSE.gbm
#GBM model with Gaussian without 9B using grade.bin
model.gbm = gbm(Number.of.Promotions
            ~X2012+X2013
            +Grade.Bin+Function+Tenure.in.Role,
            data = data.training.grade.bin,
            distribution="gaussian",
            n.trees=ntrees,
            shrinkage=0.05,
            interaction.depth=2,
            bag.fraction = 0.5,
            train.fraction = 0.5,
            cv.folds = 3)
predictions.gbm = predict(model.gbm, newdata = data.validation.grade.bin, n.trees=ntrees)
validation.residuals.gbm = data.validation.grade.bin$Number.of.Promotions-predictions.gbm
validationRMSE.gbm = sqrt(mean(validation.residuals.gbm^2, na.rm = TRUE))
rmseMatrix[i,6] = validationRMSE.gbm
#GBM model with Gaussian with 9B using grade
model.gbm = gbm(Number.of.Promotions
            ~X2012+X2013
            +X2013.Grade+Function+Tenure.in.Role+X9B,
            data = data.training.grade,
```

```
            distribution="gaussian",
            n.trees=ntrees,
            shrinkage=0.05,
            interaction.depth=2,
            bag.fraction = 0.5,
            train.fraction = 0.5,
            cv.folds = 3)
  predictions.gbm = predict(model.gbm, newdata = data.validation.grade, n.trees=ntrees)
  validation.residuals.gbm = data.validation.grade$Number.of.Promotions-predictions.gbm
  validationRMSE.gbm = sqrt(mean(validation.residuals.gbm^2, na.rm = TRUE))
  rmseMatrix[i,7] = validationRMSE.gbm
  #GBM model with Gaussian with 9B using grade.bin
  model.gbm = gbm(Number.of.Promotions
            ~X2012+X2013
            +Grade.Bin+Function+Tenure.in.Role+X9B,
            data = data.training.grade.bin,
            distribution="gaussian",
            n.trees=ntrees,
            shrinkage=0.05,
            interaction.depth=2,
            bag.fraction = 0.5,
            train.fraction = 0.5,
            cv.folds = 3)
  predictions.gbm = predict(model.gbm, newdata = data.validation.grade.bin, n.trees=ntrees)
  validation.residuals.gbm = data.validation.grade.bin$Number.of.Promotions-predictions.gbm
  validationRMSE.gbm = sqrt(mean(validation.residuals.gbm^2, na.rm = TRUE))
  rmseMatrix[i,8] = validationRMSE.gbm
}

#----------------------
#output
colnames(rmseMatrix) <- c("Linear.no9B.grade",
                    "Linear.no9B.gradebin",
                    "Linear.9B.grade",
                    "Linear.9B.gradebin",
                    "GBM.no9B.grade",
                    "GBM.no9B.gradebin",
                    "GBM.9B.grade",
                    "GBM.9B.gradebin")
rmseMatrix=as.data.frame(rmseMatrix)
summary(rmseMatrix)
#----------------------
```

```
# perform a t-test to assess whether the mean RMSE from 20-fold cross-validation without the
9B rating is higher than the mean RMSE from 20-fold cross-valudation with the 9B rating
t.test(rmseMatrix$Linear.no9B.grade, rmseMatrix$Linear.9B.grade,alternative = "g")
t.test(rmseMatrix$Linear.no9B.gradebin, rmseMatrix$Linear.9B.gradebin,alternative = "g")
t.test(rmseMatrix$GBM.no9B.grade, rmseMatrix$GBM.9B.grade,alternative = "g")
t.test(rmseMatrix$GBM.no9B.gradebin, rmseMatrix$GBM.9B.gradebin,alternative = "g")

#----------------------
#----------------------
#----------------------
#This model is a logistic regression on just 2014 promotion - yes or no
#For ease of code, will just manipulate code in Excel and save as separate csv
#Also for ease of code, removes grade.bin since results essentially the same using either exact
grade or grade.bin

#----------------------
#read data
datalog = read.csv("GILDlogistic.csv")
head(datalog)

#data cleanup
datalog$Days.to.Promo.1 = NULL
datalog$Days.to.Promo.2 = NULL
datalog$X2014 = NULL
datalog$X2015 = NULL
datalog$Number.of.Promotions = NULL
datalog$Grade.Bin = NULL
datalog$X2013.Grade = as.factor(datalog$X2013.Grade)  #make the grade a factor (since
non-linear with the jump in level)
datalog$Function = as.factor(datalog$Function)  #make the function a factor
head(datalog)

#----------------------
#randomizing the data
m = nrow(datalog)
datalog.randomized = datalog[sample(m),]

#----------------------
#cross-validation and matrix setup
aucMatrixlog = matrix(0,nrow=nFold,ncol=2)

#----------------------
```

```
#This model does not transform data set, excludes annual performance rankings from years
2014-15
logflds = createFolds(seq(1,nrow(datalog.randomized)), nFold, list = TRUE, returnTrain =
FALSE)
for (i in 1:nFold){
  ind = logflds[[i]]
  datalog.training = datalog.randomized[-ind,]
  datalog.training.no9B = datalog.training
  datalog.training.no9B$X9B = NULL
  datalog.validation = datalog.randomized[ind,]
  datalog.validation.no9B = datalog.validation
  datalog.validation.no9B$X9B = NULL
  #Logistic GBM model without 9B using grade
  model.gbmlog.no9B = buildModel("X2014.Promo", predictors=c(), datalog.training.no9B, type
='classify', method = 'GBM')
  predictions.gbmlog.no9B = genPred(model.gbmlog.no9B, newdata=datalog.validation.no9B,
method='GBM')
  aucMatrixlog[i,1] = auc(datalog.validation.no9B$X2014.Promo, predictions.gbmlog.no9B)
  #Logistic GBM model with 9B using grade
  model.gbmlog = buildModel("X2014.Promo", predictors=c(), datalog.training, type ='classify',
method = 'GBM')
  predictions.gbmlog = genPred(model.gbmlog, newdata=datalog.validation, method='GBM')
  aucMatrixlog[i,2] = auc(datalog.validation$X2014.Promo, predictions.gbmlog)
}

#----------------------
#output
colnames(aucMatrixlog) <- c("GBM.no9B.grade",
                            "GBM.9B.grade")
aucMatrixlog=as.data.frame(aucMatrixlog)
summary(aucMatrixlog)

# perform a t-test to assess whether the mean AUC from 20-fold cross-validation without the 9B
rating is lower than the mean AUC from 20-fold cross-valudation with the 9B rating
t.test(aucMatrixlog$GBM.no9B.grade, aucMatrixlog$GBM.9B.grade, alternative = 'l')
```